

# Rebecca Jennings

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## Education

**B.S. Computer Science** - University of California, Riverside

Fall 2023 - June 2027

- Relevant Coursework: Machine Organization & Assembly, Data Structures and Algorithms, Software Construction Logic Design, Principles of Web Development, Design & Architecture of Computer Systems, Discrete Structures

## Skills

**Languages:** C++, HTML, CSS, JavaScript, Python | **Frameworks:** React.js, Next.js, Tailwind CSS, Node.js, Vue.js

**Tools & Platforms:** Git, GitHub, Figma, Vercel, SolidWorks | **Soft:** Mentorship, Cross-Functional Collaboration, Prototyping, Scope Management

## Work Experience

**Software Engineer Volunteer** - Engineering for Good (EFG), Riverside, CA

August 2025 – Present

- Research and evaluate **AI platforms** for automated infographic generation, identifying effective solutions for content creation.
- Implementing a **Google Gemini API** connection and developing a service to generate infographics from prompt text, enabling **automated content creation** with user interaction.
- Testing and validating **AI-generated outputs**, integrating the service into the multiple platforms and ensuring reliability for reuse.

**Peer Academic Advisor** - BCOE, Riverside, CA

January 2025 – Present

- Serve as the first point of contact for students and faculty, delivering excellent customer service and resolving inquiries efficiently.
- Assist the team during quarterly drop-in hours by managing check-ins, assessing advising needs, and directing students to resources.
- Present guidance on course registration, student success, and engagement, while updating materials to address common advising questions.

## Projects

**R'Accessible** — Rose Hack 2026, UCR

January 2025

- Developed a web app enabling users to report accessibility barriers with **geolocation data** stored in Google Sheets via **Apps Script**.
- Built an interactive map using **Leaflet** and **OpenStreetMap** with marker clustering, dynamic zoom, and **CSV/JSON data integration**.
- Designed with accessibility best practices, including semantic **HTML**, keyboard navigation, and screen reader support.

**EcoSort** — Cutie Hack 2025, UCR

November 2025

- Architected a waste classification system using a **CNN-based model** to automate sorting for compost, recycling, and landfill streams.
- Optimized classification reliability and visual processing by implementing **OpenCV Region-of-Interest (ROI)** extraction and a **NumPy-based prediction buffer** to eliminate background noise and stabilize real-time inference.
- Engineered a seamless **software-to-hardware pipeline** by developing a **PySerial communication bridge** that translates **ML predictions** into actionable string commands for real-time mechanical bin actuation.

**Micro-blogging Website** — Principle of Web Development Course, UCR

June 2025 – July 2025

- Developed a full-featured microblogging website with **Vue.js** and **JavaScript** enabling account creation, posting, following, user feeds, and profile management using **Firestore** for data storage and retrieval.
- Implemented **collaborative folder features** allowing users to save posts, invite others to contribute, and engage in shared discussions, enhancing content exchange and user interaction.
- Aligned project design with technology principles, **CSS**, **HTML**, and by incorporating **reception, marketization, moderation, and preservation** to emphasize thoughtful user engagement.

**Playlist Organizer** – Software Construction Course, UCR

October 2024 – December 2024

- Architected a modular, **terminal-based system** in **C++** implementing custom **data structures** for user authentication and dynamic playlist management, prioritizing efficiency for search and filtering.
- **Achieved 100% test coverage** by designing a robust suite of unit tests using **Google Test**, while utilizing **Valgrind** to identify and resolve memory leaks, ensuring a memory-safe and stable execution environment.
- Facilitated an **agile/scrum** workflow by leading sprint reviews and designing **UML class diagrams** to ensure a scalable, object-oriented architecture that adhered to **SOLID software construction principles**.

## Leadership Experience

**Web Master** - SWE, UCR

April 2025 – Present

- Reviving the website by creating a maintenance guide, redesigning pages, improving usability, and developing with **CSS & HTML**.
- Supporting the marketing committee by contributing to 10+ events, ensuring smooth execution and alignment with organizational goals.
- Collaborating in weekly meetings with members to track progress, integrate feedback, and maintain consistent communication across.

**Operations** - WINC, UCR

April 2025 – Present

- Preparing presentations and coordinating room reservation while streamlining logistics and supporting execution of events and workshops.
- Managing the calendar with 30+ events while resolving scheduling conflicts among 15+ organizations, improving inter-club coordination.
- Creating weekly newsletters and collected 200+ email contacts, enhancing promotion of events and increasing participant turnout.